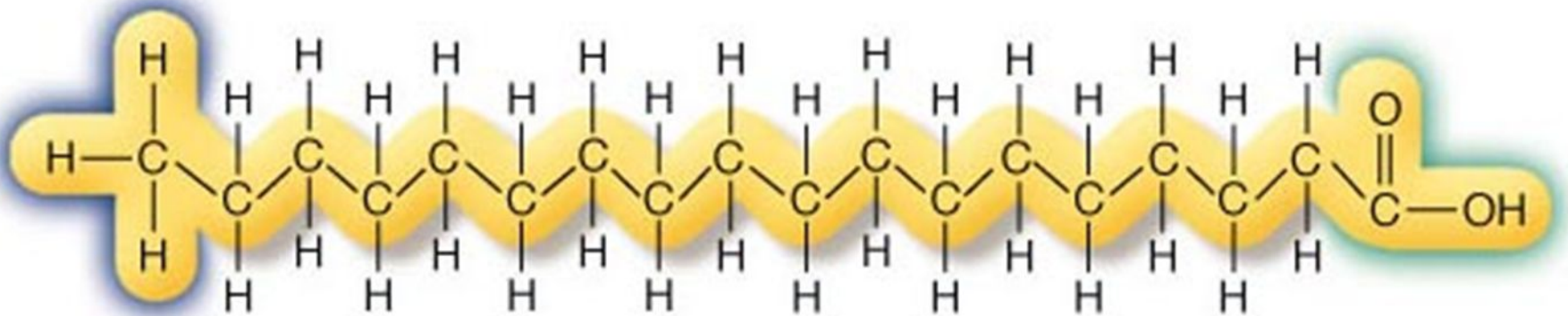
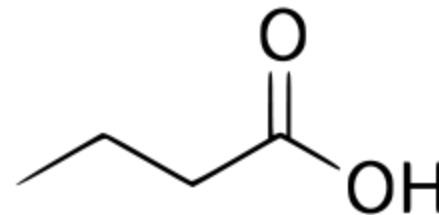


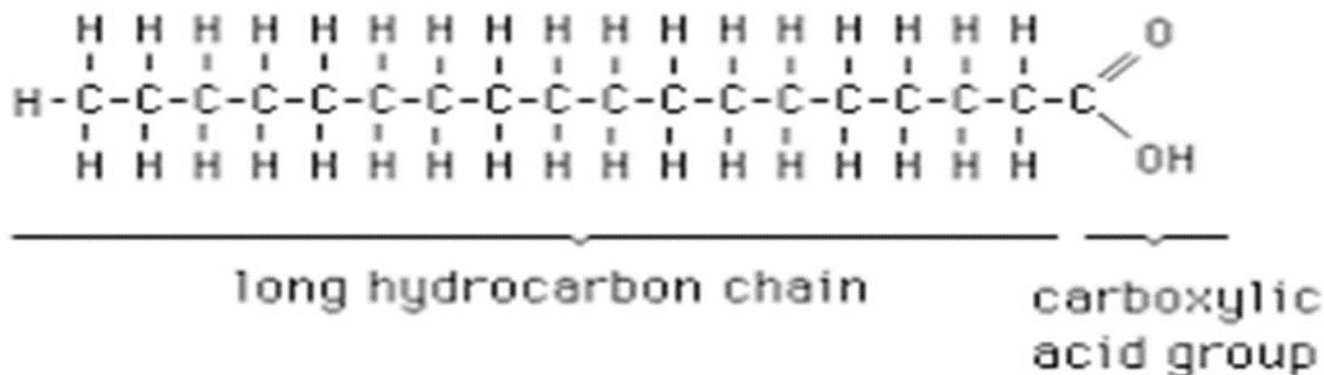
FATTY ACIDS



FATTY ACIDS (FA)



- A **fatty acid** is a monocarboxylic acid with a long hydrocarbon chain, which is either saturated or unsaturated.
- The general molecular formula of $\text{CH}_3(\text{CH}_2)_n\text{COOH}$ where n is the number of carbon atoms in the hydrocarbon chain.
- Most naturally occurring fatty acids have a chain of an even number of carbon atoms.



cont..

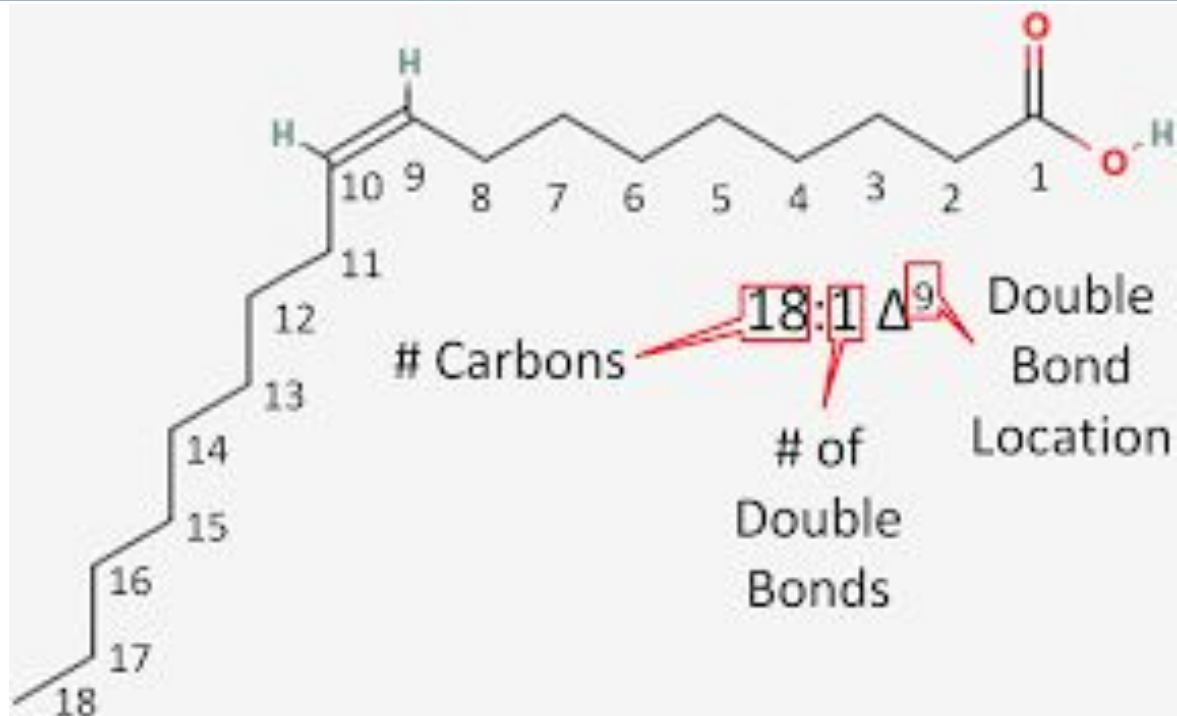
- Simplest form of lipids.
- FA are included in the group of derived lipids.
- Most common component of lipids in the body.
- Free FA are formed only during metabolism.
- Fatty acids are usually derived from triglycerides or phospholipids on hydrolysis.

Even & Odd carbon FA

- Fatty acids that occur in natural lipids are of even carbons (usually 14C – 20C)
- This is due to biosynthesis of fatty acids mainly occurs with the sequential addition of 2 carbon units.
- **Palmitic acid (16C) and Stearic acid (18C)** are most common.
- Propionic acid (3C) and valeric acid (5C) are common odd chain fatty acids.

Numbering

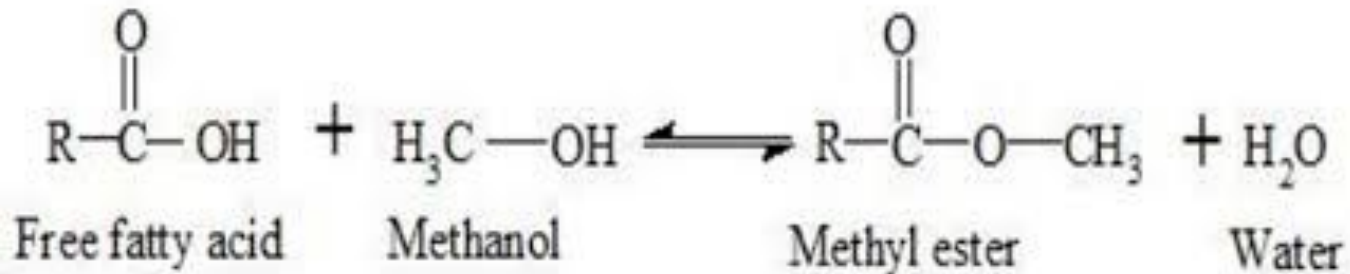
- The Carbon atoms are numbered from COOH group as 1.
- The carbons adjacent to this are 2,3,4 etc
- Last carbon atom or CH₃ group is the ω carbon.
- The omega carbon (ω) in a fatty acid is the final methyl carbon at the opposite end of the carboxylic acid group.
- Numbering from this terminal carbon determines the omega position (e.g., ω -3, ω -6) based on where the first double bond is located.



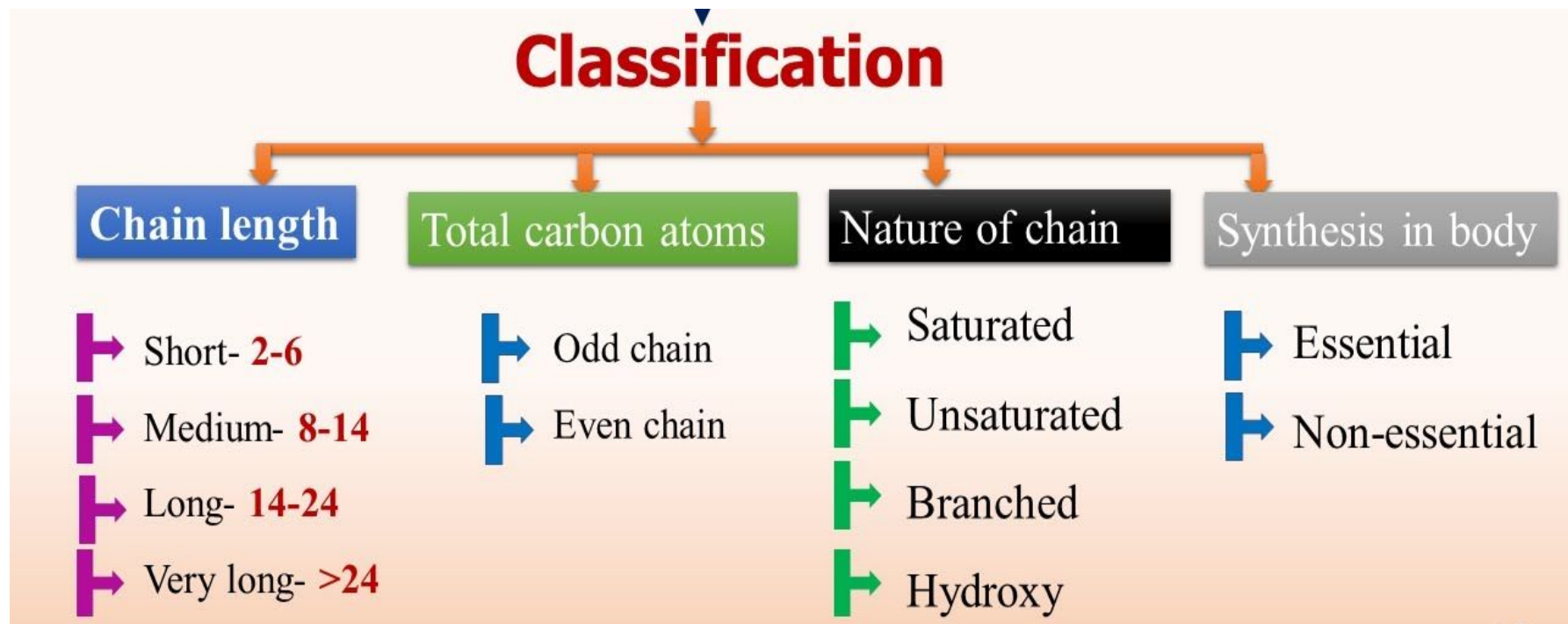
Fatty Acid with 18 carbons with a double bond at the ninth carbon atom. This is also an Omega 9 fatty acid

FORMS OF FATTY ACIDS:

1. **FREE UN-STERIFIED (IN PLASMA WITH ALBUMIN):** also known as non-esterified fatty acids (NEFA), are the active, mobilized form used for immediate energy.
2. **ESTERIFIED FATTY ACIDS:** primarily triglycerides and y.



Classification of FA on various basis



Classification of fatty acids based on length of hydrophobic chain

9

- Short chain-with 2-6 carbon atoms.
- Medium chain- with 8-12 carbon atoms.
- Long chain- with > 12 carbon atoms.

Classification of fatty acids according to degree of saturation

- Depending on double bonds
 - Saturated and
 - Unsaturated

- **Unsaturated fatty acids:**
 - MONO UNSATURATED: one double bond e:g Oleic acid
 - POLYUNSATURATED: e:g Linoleic acid

TYPES OF FATTY ACIDS

(according to the number of double bonds)



Saturated (No bond)



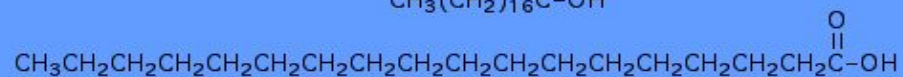
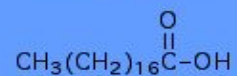
Monounsaturated (1 bond)



Polyunsaturated (>1 bond)

Saturated Fatty Acid

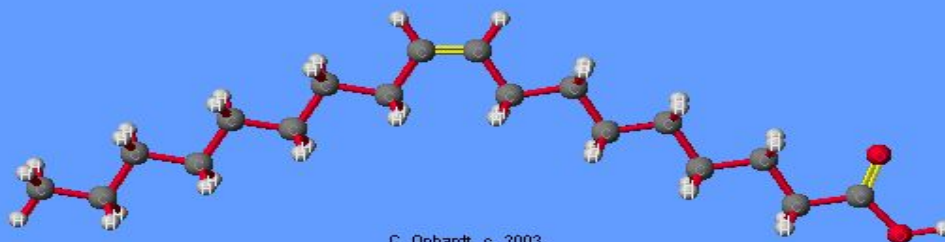
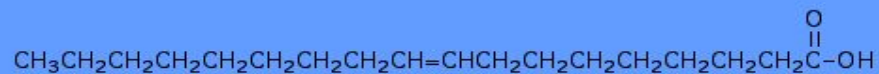
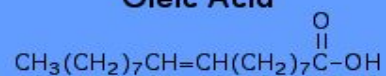
Stearic Acid



C. Ophardt, c. 2003

Unsaturated Fatty Acid

Oleic Acid



C. Ophardt, c. 2003

Saturated fatty acids:

- Are long-chain carboxylic acids that usually have between 16 and 18 carbon atoms and have no double bonds.

- Palmitic (16 C)



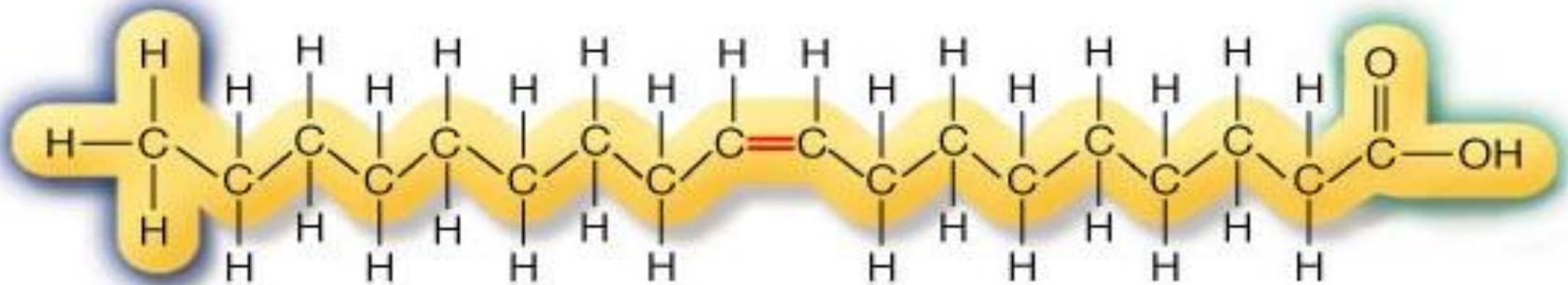
- Stearic acid (18 C)



Monounsaturated Fatty Acid (MUFA)

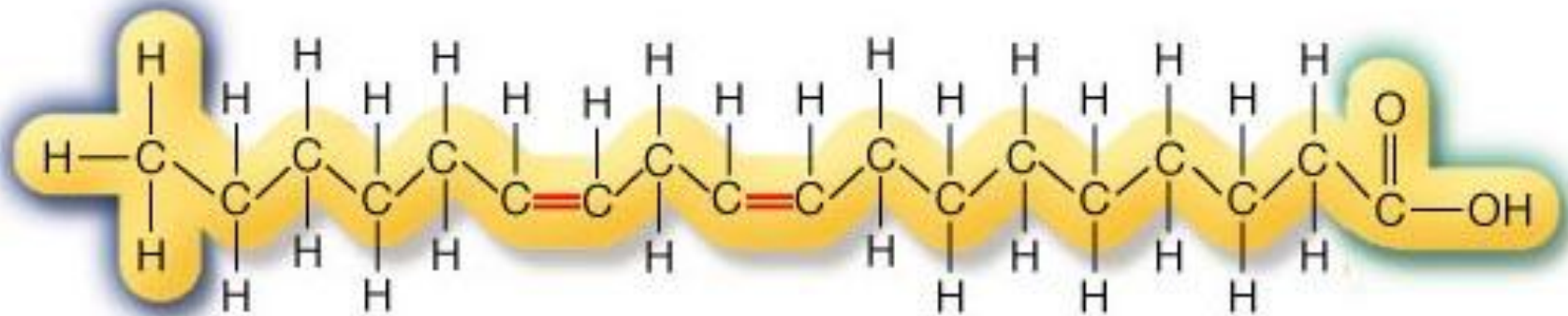
They contain one double bond

e.g. Oleic acid (18: 1; 9) ω 9



Polyunsaturated Fatty Acid (PUFA)

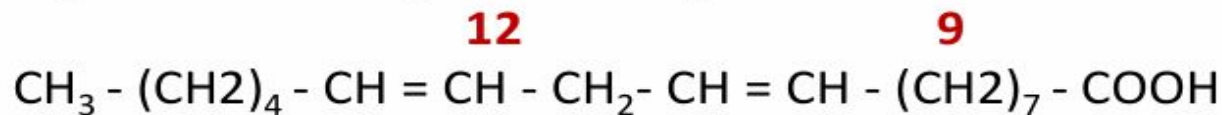
They contain more than one double bond



Polyunsaturated Fatty Acids

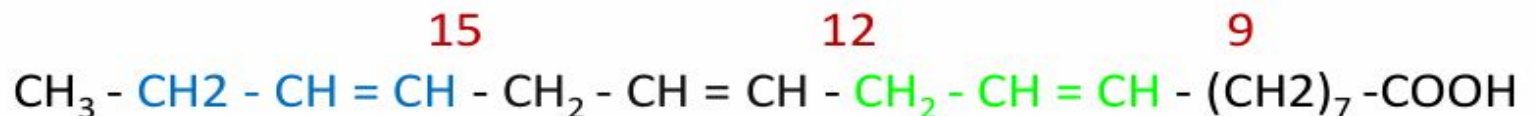
1. Dienoic fatty acids (contain 2 double bonds)

e.g. linoleic acid (18: 2 ; 9,12) ω6.



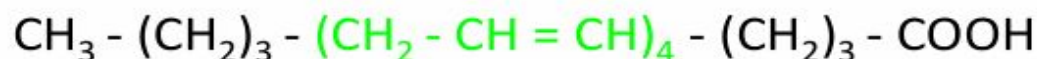
2. Trienoic fatty acids (contain 3 double bonds)

e.g. α-Linolenic (18: 3; 9,12, 15) ω3

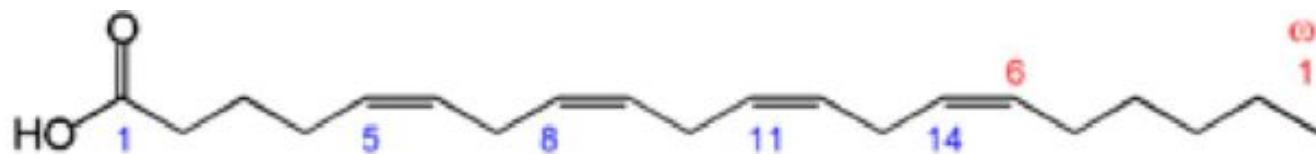


3. Tetraenoic fatty acids (contain 4 double bonds)

e.g. Arachidonic (20: 4; 5, 8, 11, 14) ω 6



Exercise: Identify the FA?

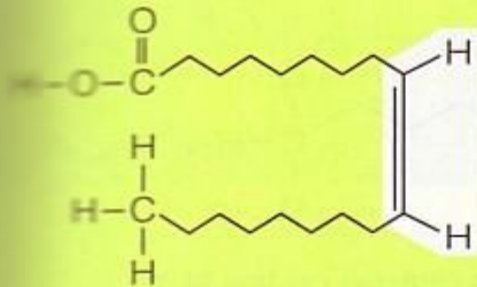


Cis-fatty acid: H's on same side of the double bond; fold into a U-like formation; naturally occurring.

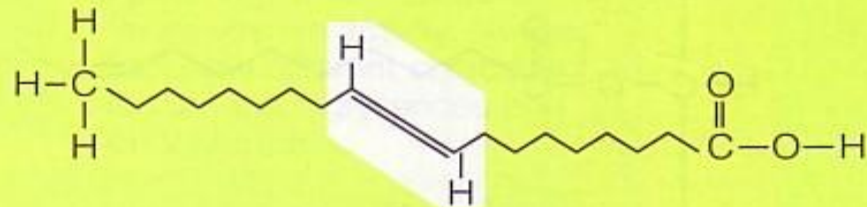
Trans-fatty acid: H's on opposite side of double bond; more linear; occur in partially hydrogenated foods.

cis- and *trans*-Fatty Acids Compared

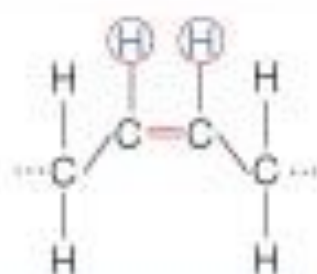
Manufacturers rarely use total hydrogenation; most often a fat is partially hydrogenated, yielding a *trans*-monounsaturated fatty acid. This example shows the *cis* configuration for the 18-carbon monounsaturated fatty acid (oleic acid) and its corresponding *trans* configuration (elaidic acid).



cis-fatty acid



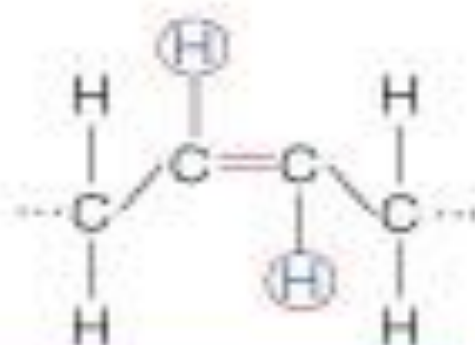
trans-fatty acid



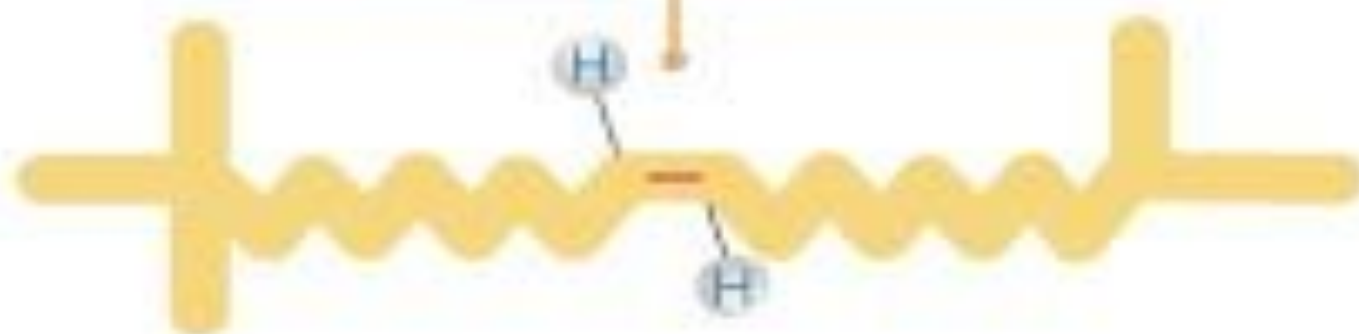
These two neighboring hydrogens repel each other, causing the carbon chain to bend



Cis form (bent)



These two hydrogens are already
as far apart as they can get



Trans form (straighter)

SATURATED FATTY ACIDS

Number of C atoms	Common Name	Systemic Name	Formula
2	Acetic acid	Ethanoic acid	CH_3COOH
4	Butyric acid	Butanoic acid	$\text{CH}_3(\text{CH}_2)_2\text{COOH}$
6	Caproic acid	Hexanoic acid	$\text{CH}_3(\text{CH}_2)_4\text{COOH}$
8	Caprylic acid	Octanoic acid	$\text{CH}_3(\text{CH}_2)_6\text{COOH}$
10	Capric acid	Decanoic acid	$\text{CH}_3(\text{CH}_2)_8\text{COOH}$
12	Lauric acid	Dodecanoic acid	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$
14	Myristic acid	Tetradecanoic acid	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$
16	Palmitic acid	Hexadecanoic acid	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$
18	Stearic acid	Octadecanoic acid	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
20	Arachidic acid	Eicosanoic acid	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$
22	Behenic acid	Docosanoic acid	$\text{CH}_3(\text{CH}_2)_{20}\text{COOH}$

UNSATURATED FATTY ACIDS

S.No.	Number of C atoms, number and location of double bonds	Family	Common Name	Systemic Name
[A]	Monoenoic acids (one double bond)			
1.	16:1;9	ω 7	Palmitoleic acid	<i>cis</i> -9-Hexadecenoic
2.	18:1;9	ω 9	Oleic Acid	<i>cis</i> -9-Octadecenoic
3.	18:1;9	ω 9	Elaidic acid	<i>trans</i> 9-Octadecanoic
[B]	Dienoic acids (two double bonds)			
1.	18:2;9,12	ω 6	Linoleic acid	all- <i>cis</i> -9,12-Octadecadienoic

UNSATURATED FATTY ACIDS(CONTD.)

S.No.	Number of C atoms, number and location of double bonds	Family	Common Name	Systemic Name
[c]	Trienoic acids (three double bonds)			
1.	18:3;6,9,12	ω 6	γ - Linolenic acid	all- <i>cis</i> -6,9,12-Octadecatrienoic
2.	18:3;9,12,15	ω 3	α -Linolenic	all- <i>cis</i> -9,12,15Octadecatrienoic
[D]	Tetraenoic acid(Four double bonds)			
	20:4;5,8,11,14	ω 6	Arachidonic	all- <i>cis</i> -5,8,11,14-Eicosatetraenoic

BIOLOGICAL IMPORTANCE OF FATTY ACIDS-

1-Fatty acids are the **building blocks of dietary fats**. The human body stores such fats in the form of **triglycerides**.

2)- Fatty acids are also **required for the formation of membrane lipids such as phospholipids and glycolipids**.

3) -They are required for the **esterification of cholesterol** to form cholesteryl esters.

4) They **act as fuel molecules** and are oxidized to produce energy.

Classification of FA according to Biological Value

ESSENTIAL FATTY ACIDS

- The FA that cannot be synthesized by the body, should be supplied through diet known as essential fatty acids.
- They are polyunsaturated fatty acids, **linoleic acid and linolenic acid**.
- Arachidonic acid is synthesized in the body from linoleic acid, so it is a non-essential fatty acid.
- Key sources include fatty fish (salmon, mackerel), nuts (walnuts), seeds (flaxseed, chia), and plant oils (soybean, canola).

Some EFA:

18:2 cisD^{9,12} linoleic acid

18:3 cisD^{9,12,15} α-linolenic acid

OMEGA-3 RICH FOODS



flax seeds
85 g/2000 cal



caviar
52 g/2000 cal



cod liver oil
44 g/2000 cal



hemp seeds
34 g/2000 cal



salmon
29 g/2000 cal



walnuts
28 g/2000 cal



oyster
20 g/2000 cal



mackerel
19 g/2000 cal



tuna
19 g/2000 cal



squid
11 g/2000 cal



mussels
11 g/2000 cal



natto
7 g/2000 cal

Non Essential Fatty Acids:

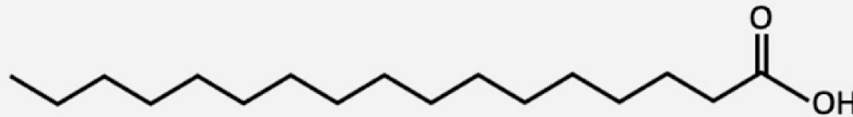


- They can be synthesized by human body, so it is not essential to take them in diet.
- They include saturated fatty acids and mono unsaturated fatty acids.

Branched Fatty Acids

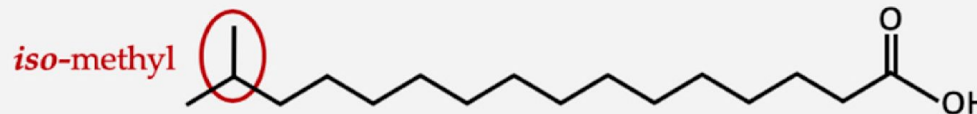
- Branched-chain fatty acids (BCFAs) are primarily saturated fatty acids containing one or more methyl branches along their carbon chain, rather than a straight chain.

Straight-chain
fatty acid



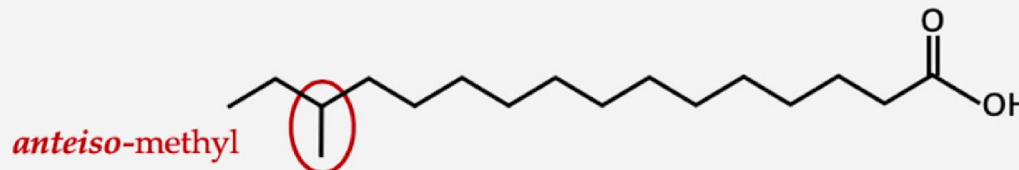
17:0

Branched-chain
fatty acid



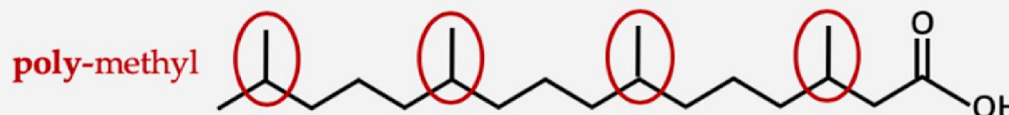
iso-17:0

15-Methylhexadecanoic acid



anteiso-17:0

14-Methylhexadecanoic acid

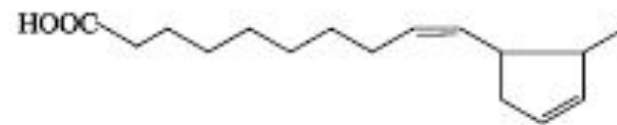
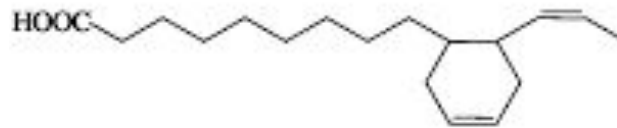


Phytanic acid

3,7,11,15-Tetramethylhexadecanoic acid

Cyclic Fatty Acids

- Cyclic fatty acids are a specialized class of fatty acids containing a carbocyclic ring in their structure.



SIGNIFICANCE OF ESSENTIAL FA:

- ❑ Components of cell membranes, structural elements of gonads and mitochondrial membrane.
- ❑ Required for brain growth and development.
- ❑ Precursors of Eicosanoids, required for the synthesis of prostaglandins.



cont..

- ❑ They have a cardio protective role- Lower serum cholesterol and increase HDL levels.
- ❑ Prevent fatty liver formation.
- ❑ Linolenic acid is necessary for optimal vision

Cont..

- **Deficiencies of essential polyunsaturated fatty acids may cause a wide variety of symptoms, including retarded growth in children, reduced fertility and pathologic changes in the skin.**

What causes Phrynoderma?

Phrynoderma is associated with nutritional deficiency, particularly:

1. Vitamin A deficiency (commonly).
2. Severe malnutrition.
3. Vitamin E and B deficiency.
4. Essential fatty acid deficiency.





THANK YOU